Project: To-Do List API **CR ID:** CR-003 — Scheduled Overdue Tasks **All statuses below are initial. No test has been executed.**

Test ID	Requirement ID	Test Scenario	Expected Result	Status
TC-003-01	FR-02	Call detection directly with a dataset containing a task whose `dueDate` is yesterday and `status = IN_PROGRESS`	Task is included in the overdue result	Not Executed
TC-003-02	FR-02	Call detection with a task whose `dueDate` is today	Task is **not** included (today is not overdue, per the business rule)	Not Executed
TC-003-03	FR-02	Call detection with a task whose `dueDate` is yesterday but `status = DONE`	Task is **not** included	Not Executed
TC-003-04	FR-02	Call detection with a task whose `dueDate` is in the future	Task is **not** included	Not Executed
TC-003-05	FR-02	Call detection with a task that has no `dueDate` set	Task is **not** included (no due date means nothing to be overdue against)	Not Executed
TC-003-06	FR-05	Invoke `OverdueTaskService.detectOverdueTasks(date)` directly in a unit test, without the scheduler firing	Method executes and returns correct results independent of any cron trigger	Not Executed
TC-003-07	FR-01	Change the configured cron expression and restart	Job fires on the new schedule instead of the old one (verified via log timestamps or a test-scoped trigger override)	Not Executed
TC-003-08	FR-04	Run detection against a dataset with 3 overdue tasks	Log output records a summary showing count = 3 and the matching task ids	Not Executed
TC-003-09	FR-03	Run detection against a task that is overdue	After the run, the task's `status` and `dueDate` in the database are unchanged (detection is read-only)	Not Executed
TC-003-10	FR-06, FR-07 *(Option B only — conditional on scope approval)*	Run detection twice in a row against the same unchanged overdue task	Exactly one `TaskNotification` row exists after both runs, not two	Not Executed
TC-003-11	FR-06, FR-07 *(Option B only)*	Change an already-notified task's `dueDate` to a new past date, then run detection again	A second, distinct notification is created for the new due date (not treated as a duplicate)	Not Executed
TC-003-12	— (error handling)	Simulate a transient exception inside the detection logic during a scheduled run	Exception is logged; the application keeps running and the next scheduled run still fires	Not Executed

Notes

- TC-003-10 and TC-003-11 apply only if Option B (notification persistence) is approved by the project owner; otherwise they are not applicable and should be removed from the executed test plan.
- TC-003-07 may be implemented by exposing the cron property as an externally overridable Spring property in the test profile rather than waiting for a real cron interval to elapse.